

1) What are the three measures of central tendency?

Ans : mean, Median, mode

2) What is the difference between the mean, median, and mode? How are they used to measure the central tendency of a dataset?

Ans : mean find average

Median find middle value

Mode find most frequently value

3) Measure the three measures of central tendency for the given height data:

[178,177,176,177,178.2,178,175,179,180,175,178.9,176.2,177,172.5,178,176.5]

Ans : mean :- 177.01

Median :- 177

Mode :- 178, 177

4) Find the standard deviation for the given data:

[178,177,176,177,178.2,178,175,179,180,175,178.9,176.2,177,172.5,178,176.5]

Ans : mean :- 177.01

Variance :- 3.20

Std :- 1.79

5) How are measures of dispersion such as range, variance, and standard deviation used to describe the spread of a dataset? Provide an example.

Ans : range simple spread data $180 - 172.5 = 7.5$

Variance Squared spread 3.20

Standard deviation Actual spread from mean 1.79

6) What is a Venn diagram?

Ans :

7) For the two given sets $A = (2,3,4,5,6,7)$ & $B = (0,2,6,8,10)$. Find:

(i) $A \cap B$

(ii) $A \cup B$

Ans : $A \cap B = (2,6)$

$A \cup B = (0,2,3,4,5,6,7,8,10)$

8) What do you understand about skewness in data?

Ans : skewness describe the shape of a data distribution

Symmetrical Distribution is perfect data

Positive Skewed (Right skewed)

Negatively Skewed (left Skewed)

9) If a data is right skewed then what will be the position of median with respect to mean?

Ans : $\text{Mean} > \text{Median} > \text{Mode}$

10) Explain the difference between covariance and correlation. How are these measures used in statistical analysis?

Ans : **covariance** show direction only

Correlation show direction + strength

Find relation between variables

11) What is the formula for calculating the sample mean? Provide an example calculation for a dataset.

Ans : formula $\bar{x} = \sum x/n$

E.x. $10+12+15+17+20 = 74$

$74/5 = 14.8$

12) For a normal distribution data what is the relationship between its measure of central tendency?

Ans : normal distribution bell-shaped curve

Measure of central tendency mean=median=mode

13) How is covariance different from correlation?

Ans : **covariance** show direction only

Correlation show direction + strength

14) How do outliers affect measures of central tendency and dispersion? Provide an example

Ans : mean without outliers data

E.x. $2+4+6+8+10=30/5 = 6$

Mean with outliers data

E.x. $1+3+5+7+50=66/5=13.2$

Median without outliers data

E.x. 2,3,4,5,6 = 4

Median with outliers data

E.x. 2,3,4,5,6,100 = $(4+5)/2 = 4.5$

Mode is most frequent value outliers not affect mode

Measures of dispersion

Range without outliers data

E.x. 2,3,4,5,6 = $6-2 = 4$

Range with outliers data

E.x. 2,3,4,5,6,100 = $100-2 = 98$

Variance without outliers data

E.x. $2+4+6+8+10= 30/5 = 6$

X mean = -4,-2,0,2,4

Variance = $X^2 = 16+4+0+4+16 = 40/5 = 8$

Standard deviation = square root of variance is std = 2.83